

Case Study: BookNest – A Mobile Library App

Chaya Public Library wants to create a new mobile app called BookNest. This app will help library members borrow e-books online without visiting the library. Users must be able to sign up and log in using their national ID or library card number.

After logging in, users can search for e-books, borrow books for 14 days, and return or renew them using the app. The app should also show book details, such as title, author, summary, and availability. It should suggest books based on what the user has read before.

While reading an e-book, users should be able to bookmark pages and write personal notes. After finishing a book, they can rate and write reviews to help other readers.

To help users manage their books, the system must automatically send notifications when a book they reserved is available.

The app must support Thai and English languages. It should also have a text-to-speech feature for people who have difficulty reading.

The app should be easy to use, especially for elderly users. It must work on both Android and iOS phones. Each screen must load in less than 2 seconds and support up to 1,000 users at the same time during busy hours.

The app must protect user data and follow Thai PDPA laws. All personal information must be encrypted using AES 256-bit encryption.

The development team must use React Native to build the app. The app must be completed within 4 months, with a total cost not more than 350,000 Baht.

1) Functional Requirements (what the system must do)

- Account & access
 - Users can **sign up** and **log in** via **national ID** or **library card number**.
 - Users can **log out** and recover access (e.g., reset PIN/password).
- Catalog & discovery
 - Users can **search** for e-books (by title/author/keywords).
 - App displays **book details**: title, author, summary, availability.
 - System provides **personalized recommendations** based on reading history.
- Borrowing lifecycle
 - Users can **borrow** e-books for **14 days**.
 - Users can **return** e-books via the app.
 - Users can **renew** loans (subject to policy/availability).
 - Users can **reserve/hold** unavailable e-books.
 - System **automatically notifies** users when a reserved book becomes available.
- Reading experience
 - In-app e-book reader supports **bookmarks** and **personal notes**.
 - **Text-to-speech** is available while reading.
- Reviews & ratings
 - Users can **rate** and **write reviews** after finishing a book; reviews are viewable by others.
- Internationalization & accessibility

- App supports **Thai and English** UI.
- **Accessibility** features for elderly users (e.g., larger fonts, high contrast, simple flows).

2) Non-Functional Requirements (how well the system must perform)

- **Performance**
 - **Each screen loads < 2 seconds** on typical 4G/Wi-Fi.
 - Supports **≥ 1,000 concurrent users** during peak hours.
- **Usability & accessibility**
 - Interfaces are **simple and senior-friendly** (large touch targets, readable typography).
 - **Text-to-speech** quality is intelligible; language follows device locale or in-app setting.
- **Security & privacy**
 - Complies with **Thai PDPA**.
 - **All personal data encrypted with AES-256** at rest; secure transport (TLS 1.2+).
 - Role-appropriate access to user and borrowing data; secure session management.
- **Compatibility/portability**
 - Runs on **Android and iOS** (current + previous major versions).
- **Reliability/availability**
 - Service uptime target (e.g., **≥ 99.5% monthly**).
 - Data backup & restore for accounts, notes, bookmarks, reviews (e.g., **daily backups**, 30-day retention).
- **Maintainability & observability**
 - Error logging, crash analytics, and basic usage metrics in place.
 - Modular codebase to support new formats/features.
- **Internationalization**
 - All user-facing strings externalized; **Thai/English** parity.

3) Domain Requirements (library-specific business rules & policies)

- **Identity sources** are **national ID** and **library card** (library membership model).
- **Loan period is 14 days** per e-book.
- **Renewal rules:** renewal permitted if no active reservations (policy to be parameterized).
- **Reservation (hold) queue** with **FIFO** priority; auto-notification when available.
- **Availability** reflects current licenses/copies (DRM/license count).
- **Reviews & ratings** are associated with library user accounts; moderation policy may apply.
- **Book metadata** structure: title, author, summary, availability (minimum fields), extensible to genre, ISBN, cover image, etc.
- **Reading history** is retained to power recommendations (with user consent per PDPA).

4) Constraints (hard limits/mandates the project must obey)

- **Technology constraint:** Must be built with **React Native**.

- **Platform constraint:** Must run on **Android and iOS**.
- **Regulatory constraint:** Must **comply with Thai PDPA**.
- **Cryptography constraint:** **AES-256** for personal data at rest.
- **Time constraint:** **Complete within 4 months**.
- **Budget constraint:** **≤ 350,000 THB** total cost.

Case Study: ParkEasy – A Smart Parking App

The city of Chanthaburi has many problems with traffic and parking. Many people waste time looking for a place to park. To solve this, the city government wants to create a mobile app called ParkEasy. This app will show available parking spots in real time, help users book a space, and guide them with maps.

The app must also support payment for parking fees using mobile banking or QR code. After users park, they can check how much time is left or extend their time through the app.

The system should work with sensors installed in public parking areas. When a car enters or leaves, the system updates the parking status. A web dashboard will help parking officers monitor all locations.

The city will also invite private parking lots (like shopping malls) to join the platform. Each parking area manager can update space availability and prices. Citizens can rate parking areas and leave feedback.

The government plans to make this app free for citizens and hopes it will reduce traffic problems in the city.

1. Primary Stakeholders (Direct Users)

- **Citizens / Drivers (End Users)**

Role: Use the mobile app to find available parking spaces, book a spot, pay parking fees, extend parking time, and give ratings/feedback.

Interest: Convenient parking, reduced time searching for spaces, secure payments.

- **Parking Officers (City Staff)**

Role: Monitor parking status through the web dashboard, oversee sensor system operations, enforce parking rules.

Interest: Efficient monitoring, reduced manual work, reliable real-time updates.

- **Private Parking Lot Managers (e.g., malls, offices, hospitals)**

Role: Manage availability, update pricing, ensure their parking spots are visible in the app.

Interest: Attract more users, increase revenue, optimize lot usage.

2. Secondary Stakeholders (Supporting Roles)

- **City Government (Chanthaburi Municipality)**

Role: Project owner, funding and policy maker, ensures system integration with public infrastructure.

Interest: Reduce traffic congestion, improve citizen satisfaction, promote smart city initiatives.

- **App Developers / IT Vendor**

Role: Design, develop, test, and maintain the mobile app and backend system (including payment gateway and sensor integration).

Interest: Deliver system within scope, budget, and time; ensure maintainability and performance.

- **Payment Service Providers (Mobile Banking, QR Code platforms)**
Role: Provide secure transactions and integrate payment APIs with the app.
Interest: Expand user base, ensure transaction security and compliance.
 - **Sensor Providers / IoT Vendors**
Role: Install and maintain parking sensors; provide accurate real-time data feeds to the app.
Interest: Reliable technology adoption and long-term maintenance contracts.
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3. External Stakeholders

- **Local Businesses (near parking areas)**
Role: Indirect beneficiaries since better parking access may increase customer visits.
Interest: Improved customer flow, fewer traffic issues.
- **Regulators / Legal Authorities**
Role: Ensure compliance with financial regulations, data privacy (e.g., PDPA in Thailand), and traffic laws.
Interest: Legal compliance, consumer protection, secure handling of personal and payment data.

Case Study: StudyMate – A Study Group Finder App

Many university students have trouble studying alone, especially before exams. Some students don't know who else is studying the same subject or where to find study partners. Others want to join a study group but don't know how.

A team of computer science students wants to build a mobile app called StudyMate. The app will help students find and join study groups based on their courses and interests. Students can create a study group, invite others, or search for public groups in their subject area. They can also chat, set meeting times, and share documents in the group.

The team wants this app to be simple, fast, and useful for all students. It will be available in both Android and iOS. The goal is to help students learn better together and reduce exam stress.

FOR

university students who struggle to study alone or find it hard to connect with peers in the same subject

WHO

need an easy way to discover, join, and organize study groups to prepare for exams and reduce stress

OUR PRODUCT

StudyMate – a mobile app for study group creation and collaboration

THAT

helps students find the right peers, join subject-based groups, chat, schedule meetings, and share documents so they can learn better together

UNLIKE

informal messaging groups (e.g., LINE, Facebook groups) that are unorganized and not course-specific

OUR PRODUCT

is purpose-built for academic collaboration, offering course-linked group discovery, structured scheduling, and study-focused features in a simple, fast app available on both Android and iOS

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Actors

- **Library Member** (primary)
- **ID/Library Card Verification Service** (external)
- **Push Notification Service** (external)
- **Text-to-Speech (TTS) Engine** (external)
- **E-book DRM/Content Provider** (external)

Main Use Cases (Library Member)

- Sign Up
- Log In
- Search E-books
- View Book Details
- View Recommendations
- Borrow E-book (14 days)
- Return E-book
- Renew Loan
- Reserve (Hold) E-book
- Read E-book
 - Bookmark Page (include)
 - Add Personal Note (include)
 - Text-to-Speech (include; via TTS Engine)
- Rate Book
- Write Review
- Change Language (Thai/English)

System/Service Use Cases

- Notify Availability (to Member; via Push Notification Service)
- Verify Identity (via ID/Library Card Verification Service)
- Deliver Licensed Content (via DRM/Content Provider)

Key Relationships

- **Borrow E-book** — <<include>> **Verify Identity**, <<include>> **Deliver Licensed Content**
- **Reserve E-book** — <<extend>> **Borrow E-book** (when not available)
- **Notify Availability** — <<include>> from **Reserve E-book**
- **Read E-book** — <<include>> **Bookmark Page, Add Personal Note, Text-to-Speech**
- **View Recommendations** — uses reading history (internal dependency)
- **Sign Up / Log In** — <<include>> **Verify Identity**

Example

Use case	Edit a post
Summary	Author wants to edit a post
Actors	Author, Blog Platform
Preconditions	Author has opened a browser and is currently on the Blog Platform dashboard
Basic sequence	1. Author clicks on “Posts” on the navigation menu 2. Author clicks on the post that should be edited 3. Author makes the necessary changes to the title and the content 4. Author presses the save button 5. Success message should be visible on the screen
Exceptions	Step 5: Somebody else has edited the same post before the save button was pressed. Error message should be visible saying that another user has edited the same post and give the user instruction on what to do.
Post conditions	The post has been edited successfully.

1. Sign Up

- **Summary:** A new user registers for a BookNest account.
- **Actors:** Library Member, ID/Library Card Verification Service.
- **Preconditions:** The app is installed and running; the user has a valid national ID or library card number; the device has network connectivity.
- **Basic sequence:**
 1. The member taps “Sign Up”.

2. The member enters their national ID or library card number and any required personal details.
 3. The app sends these details to the ID/Library Card Verification Service for validation.
 4. If the verification succeeds, the app creates a new account.
 5. The app displays a success message and logs the user in.
- **Exceptions:**
 - *Step 3:* If the ID is invalid or already in use, an error message prompts the user to correct the information.
 - *Network failure:* The app informs the user of connectivity issues and suggests retrying.
 - **Post conditions:** The user account is created and initialized; the user is logged in and may proceed to use other features.
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2. Log In

- **Summary:** An existing member accesses their BookNest account.
 - **Actors:** Library Member, ID/Library Card Verification Service.
 - **Preconditions:** The member has already registered; the app has network connectivity.
 - **Basic sequence:**
 1. The member taps “Log In”.
 2. The member enters their ID/library card number and authentication credentials (e.g., password or PIN).
 3. The app validates the credentials with the verification service.
 4. Upon successful authentication, the app grants access to the user’s account.
 - **Exceptions:**
 - *Step 3:* Invalid credentials cause an error message and allow retrying.
 - *Account locked:* Too many failed attempts trigger an account lock and instructions to reset credentials.
 - **Post conditions:** The member is authenticated and redirected to the main interface.
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3. Search E-books

- **Summary:** The member searches the library’s e-book catalog.
- **Actors:** Library Member, BookNest App.
- **Preconditions:** The member is logged in; the app can access the catalog database.
- **Basic sequence:**
 1. The member enters search keywords or filters (e.g., title, author, genre).
 2. The app queries the catalog and displays matching results with basic details and availability.
- **Exceptions:**
 - *Invalid search input:* If the search string is empty or uses unsupported characters, the app displays a helpful message and prompts for new input.
 - *Server unavailable:* The app shows an error and suggests retrying later.

- **Post conditions:** Search results are displayed; the member can select a result for more details.
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4. View Book Details

- **Summary:** The member examines detailed information about a chosen e-book.
 - **Actors:** Library Member.
 - **Preconditions:** The member is logged in and viewing a list of search results or recommendations.
 - **Basic sequence:**
 1. The member selects a specific e-book from a list.
 2. The app shows the title, author, summary, and availability status; it may also show cover art and other metadata.
 3. The app provides options to borrow or reserve the book (depending on availability).
 - **Exceptions:**
 - If metadata retrieval fails, a message appears indicating the book details cannot be displayed now.
 - **Post conditions:** The book's details are presented to the member; borrowing or reservation may be initiated.
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5. View Recommendations

- **Summary:** The member views personalized book suggestions based on their reading history.
 - **Actors:** Library Member.
 - **Preconditions:** The member is logged in and has reading history stored with the app.
 - **Basic sequence:**
 1. The member taps "Recommendations".
 2. The app analyzes past loans, ratings, and reviews to generate personalized suggestions.
 3. Recommended titles appear in a list with key details and an option to view full details or borrow.
 - **Exceptions:**
 - If no history exists (e.g., a new account), the app displays a message like "No recommendations yet—start reading to get suggestions."
 - **Post conditions:** A list of suggested e-books appears, enhancing discovery.
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6. Borrow E-book

- **Summary:** The member checks out an available e-book for a 14-day loan.
- **Actors:** Library Member, ID/Library Card Verification Service, E-book DRM / Content Provider.
- **Preconditions:** The member is logged in; the book is available; the member has not exceeded their borrowing limit.
- **Basic sequence:**

1. In the book details, the member taps “Borrow”.
 2. The app confirms the member’s identity via the verification service.
 3. The app processes the loan and reserves a license from the DRM/content provider.
 4. The book appears in the member’s “My Loans” with a due date set for 14 days from the current date.
 5. The app allows the member to open and read the e-book.
- **Exceptions:**
 - *Step 2:* If identity verification fails (e.g., card expired), the borrow request is denied with instructions to update membership.
 - *Step 3:* If all licenses are currently in use (despite availability status), the app offers to place a reservation instead.
 - **Post conditions:** A loan record is created with a due date; the member can access the e-book offline for the loan period.
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7. Return E-book

- **Summary:** The member returns a borrowed e-book before or on its due date.
 - **Actors:** Library Member, E-book DRM / Content Provider.
 - **Preconditions:** The member has an active loan; the e-book is still within the loan period.
 - **Basic sequence:**
 1. The member opens “My Loans” and selects the loan to return.
 2. The member taps “Return”.
 3. The app removes the e-book from the member’s loan list and frees the license with the DRM/content provider.
 4. The app displays a confirmation message.
 - **Exceptions:**
 - *Network or DRM service failure:* The return may temporarily fail; the app prompts to retry later while keeping the local file in read-only mode.
 - **Post conditions:** The loan record is closed; the e-book cannot be accessed unless borrowed again.
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8. Renew Loan

- **Summary:** The member extends the borrowing period for a currently loaned e-book.
- **Actors:** Library Member, DRM / Content Provider.
- **Preconditions:** The member has an active loan; the book has no pending reservations by other users.
- **Basic sequence:**
 1. The member selects a loan in “My Loans”.
 2. The member taps “Renew”.
 3. The app verifies that no one has reserved the book.
 4. If renewable, the due date is extended by 14 days.
 5. A confirmation message appears.

- **Exceptions:**
 - *Step 3:* If the book is reserved by another user, renewal is denied, and the member must return the book when the original due date arrives.
 - **Post conditions:** The loan record is updated with a new due date (if renewable) or remains unchanged with an explanatory message.
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9. Reserve E-book

- **Summary:** The member places a hold on a book that is currently not available.
 - **Actors:** Library Member, ID/Library Card Verification Service, Push Notification Service.
 - **Preconditions:** The member is logged in; the book has no available copies; the member has borrowing privileges.
 - **Basic sequence:**
 1. The member taps “Reserve” on a book’s detail page.
 2. The app verifies the member’s account status and eligibility (e.g., no overdue items).
 3. The book is placed in a reservation queue (first come, first served).
 4. The member receives confirmation of the reservation and an estimated wait time.
 - **Exceptions:**
 - If the member already has an active loan or reservation for the same book, the app shows a message indicating that duplicate reservations are not allowed.
 - **Post conditions:** The reservation is logged with a timestamp; the system will notify the member when it’s available.
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10. Notify Availability

- **Summary:** The system informs a member that a reserved e-book is now available to borrow.
 - **Actors:** Push Notification Service, Library Member.
 - **Preconditions:** The member has an active reservation; the reserved book’s license becomes free; push notifications are enabled.
 - **Basic sequence:**
 1. The system detects that the next copy of the reserved book is free.
 2. The system sends a push notification to the member via the Push Notification Service.
 3. The notification includes an action button (e.g., “Borrow now”) or instructs the member to open the app.
 4. The member borrows the e-book as in the Borrow use case.
 - **Exceptions:**
 - If the member’s device is offline or push notifications are disabled, the app queues a local message that appears when the app is next opened.
 - **Post conditions:** The member is informed of book availability and can choose to borrow it.
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11. Read E-book

- **Summary:** The member reads a borrowed book within the app.
 - **Actors:** Library Member, Text-to-Speech Engine.
 - **Preconditions:** The member has an active loan; the e-book file is downloaded; device supports reading.
 - **Basic sequence:**
 1. The member selects a book from “My Loans” and opens it.
 2. The e-book reader displays the content with navigation controls.
 3. The member can:
 - **Bookmark Page:** Tap a bookmark icon to mark the current page.
 - **Add Personal Note:** Highlight a section and add a textual note.
 - **Text-to-Speech:** Activate TTS to have the page read aloud via the Text-to-Speech Engine.
 4. The reader saves bookmarks and notes automatically.
 - **Exceptions:**
 - *Step 3:* If text-to-speech fails (e.g., no audio output), an error message explains the problem.
 - *File corruption:* If the e-book file is corrupted, the app prompts the user to re-download.
 - **Post conditions:** Reading progress, bookmarks, and notes are stored; the member may exit and resume later.
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12. Rate Book

- **Summary:** After finishing a book, the member assigns a numerical rating.
 - **Actors:** Library Member.
 - **Preconditions:** The member has completed or returned a loan; rating requires being logged in.
 - **Basic sequence:**
 1. The member navigates to the book’s detail page or “Completed Reads”.
 2. The member selects a star rating (e.g., 1–5 stars).
 3. The app records the rating and updates aggregated scores.
 4. The member sees a confirmation message.
 - **Exceptions:**
 - If the member hasn’t borrowed the book previously, the rating option is disabled with a tooltip explaining why.
 - **Post conditions:** The rating is stored and contributes to overall book scores.
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13. Write Review

- **Summary:** The member writes a textual review of a book they have finished.
- **Actors:** Library Member.
- **Preconditions:** The member has read the book; the rating (optional) may already have been given.
- **Basic sequence:**
 1. From the book’s detail page, the member taps “Write Review”.

2. The member enters a title and body for the review.
 3. The member submits the review.
 4. The app stores it and displays it to other users (subject to moderation policies).
- **Exceptions:**
 - Offensive or inappropriate language may be flagged and require moderation; the member is notified if the review is rejected.
 - **Post conditions:** The review is saved and associated with the member's profile; other users can read it.
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14. Change Language

- **Summary:** The member switches the app interface between Thai and English.
- **Actors:** Library Member.
- **Preconditions:** The member is logged in or using the app; the app supports multiple languages.
- **Basic sequence:**
 1. The member opens the settings menu.
 2. The member selects "Language".
 3. The member chooses Thai or English.
 4. The app reloads the UI strings in the chosen language.
- **Exceptions:**
 - If the language resource fails to load (e.g., missing files), the app defaults to English and displays a message.
- **Post conditions:** All menus, labels, and messages appear in the selected language immediately.